

Fucheng Liu (Eric)

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EDUCATION

National University of Singapore | Aug 2023 - May 2027 (expected)

B.Sc., Mathematics; Second Major in Quantitative Finance; Minor in Data Science and Analytics.

Selected coursework represented in projects: Discrete Optimisation (MA4254), Regression Analysis (ST3131), Artificial Intelligence: Technology and Impact (IT1244).

Shanghai Jiao Tong University | Summer Session, Jun - Jul 2025

Algebra and Number Theory (A); Complex Variable (A-).

RESEARCH INTERESTS

Financial time series, multimodal learning, model uncertainty, and optimization.

MANUSCRIPTS UNDER REVIEW

Fucheng Liu, Chenxue Yang. *Adaptive Temporal-Context Fusion for Policy-Aware Multimodal Agricultural Stock Movement Prediction*. Under review at *Expert Systems with Applications*, 2026.

RESEARCH EXPERIENCE

Policy-Aware Multimodal Stock Movement Prediction

- Investigated the fusion of historical price features, macroeconomic policy information, and corporate announcements for agricultural stock movement prediction.
- Contributed experimental design, implementation, debugging, comparative experiments, manuscript revision, and initial drafting.
- Used Python for data processing, model training, and evaluation; examined the role of temporal context in multimodal prediction.

SELECTED INTERNSHIP EXPERIENCE

China Merchants Securities | Quantitative Research Intern | May - Aug 2026

- Replicated and tested academic factors, including Tail Risk Exposure and Skewness Dispersion, using daily A-share data, portfolio backtesting, and risk-return evaluation.
- Studied corporate events such as private placements using average abnormal returns, hit rates, and price changes around announcements.
- Conducted multimodal prediction experiments across 2 forecasting horizons (5 and 20 trading days); used Python, Wind, and iFinD for data processing, analysis, and research exhibits.

Avalok Capital | Quantitative Research Intern (Remote) | Oct - Dec 2025

- Replicated and backtested 101 formulaic alpha signals; built and evaluated multi-factor models to study liquidity, volatility, and macro-regime effects.
- Observed backtest Sharpe ratios ranging from 0.3 to 1.5 after accounting for transaction costs on both purchases and sales; analyzed factor correlations to assess signal redundancy.

SELECTED PROJECTS

Optimization-Based Group Testing | MA4254 Discrete Optimisation

- Formulated sparse recovery in non-adaptive group testing as an integer linear program with binary variables and test-consistency constraints.
- Implemented and compared 4 recovery approaches: ILP, LP relaxation with randomized rounding, DNI, and DI.
- Compared exact recovery probability and runtime in simulations with up to 500 individuals under varying testing budgets.

HDB Resale Price Prediction | ST3131 Regression Analysis

- Analyzed 2,456 HDB resale transactions across 26 Singapore towns (July 2020), modelling log resale prices using property characteristics.
- Compared full and AIC-selected regression models; the final 6-predictor specification achieved an in-sample adjusted R-squared of 0.907 on log prices, with residual diagnostics.

DNA-Binding Protein Classification | IT1244 Artificial Intelligence

- Worked in a 3-person team on DNA-binding protein classification using 53,285 training and 13,452 test protein sequences.
- Contributed to code revision, execution, and analysis comparing 3 model families (XGBoost, CNN, Random Forest); the team reported a Random Forest test MCC of 0.4095.

ADDITIONAL EXPERIENCE

October Capital | Venture Capital Investment Intern | May - Jul 2025

- Supported due diligence for 3 companies, each with a proposed investment of RMB 35 million, through market research, financial analysis, and investment committee materials.

Volkswagen (Anhui) Automotive | Productivity Management Intern | May - Jul 2024

- Supported operational data tracking, workflow coordination, and cross-department process improvement.

SELECTED COMPETITIONS

Moomoo Options Paper Trading Competition | Jan 2026

Ranked 2nd among 5,432 participants in simulated options trading.

Optiver Trading Simulation Challenge | Apr 2025

Ranked in the top 10% among approximately 100 participants; implemented a grid trading strategy in a simulated environment.

TECHNICAL SKILLS

Programming and analysis: Python (NumPy, Pandas, Matplotlib), R, MATLAB, C++, Excel.

Financial data: Wind, Tonghuashun iFinD; time-series analysis, factor research, backtesting.

Mathematical background: Probability, stochastic processes, linear algebra, linear and nonlinear optimization, numerical methods.

Languages: English (fluent), Mandarin Chinese (native), Japanese (beginner).

SERVICE

St. Luke's Elder Care volunteer, Jan - May 2026; science communication activities with Singapore Science Centre through NUS SP1541, Jan - May 2025.